

## **Module 4: A+ Troubleshooting and Helpdesk**

### **DAY 4:-**

#### **Task 1:**

##### **RAM Reseating Task**

###### **Step 1: Power off and unplug**

Power off the PC completely and unplug it from the wall. If it's a laptop, remove the battery if possible.

###### **Step 2: Apply ESD precautions**

Touch a grounded metal surface before touching any internal components to prevent any static damage.

###### **Step 3: Open the case and locate RAM slots**

Open the case to locate the RAM slots, usually near the CPU Socket.

###### **Step 4: Remove the RAM module**

Push down on the retention clips at both ends of the slot until the module pops up slightly out.

###### **Step 5: Reinsert the RAM module**

Align the notch on the RAM module with the notch in the slot, then press down firmly and evenly until both retention clips click into place.

###### **Step 6: Power on and verify detection**

Power the PC back on and check that it boots normally and confirm the correct RAM capacity is detected in BIOS or the OS.

#### **Task 2:**

##### **GPU Troubleshooting Checklist**

###### **Step 1: Identify the Symptom**

Note the exact symptom: no display at all, artifacts/glitches on screen, crashes only during gaming, or the fan spinning but no output. This narrows down whether it's a display, driver, or hardware issue.

###### **Step 2: Check Physical Seating and Power Connection**

Power off the PC and confirm the GPU is fully seated in its PCIe slot, and that all PCIe power connectors (6/8-pin) from the PSU are firmly connected.

### **Step 3: Verify the correct display connection**

Confirm the monitor cable is plugged into the GPU's output ports, not the motherboard's rear I/O — a common mistake that causes 'no display' even with a working GPU.

### **Step 4: Check POST behavior**

Boot the PC and listen/watch for POST completion (beep or diagnostic LED). If POST fails specifically at the video stage, this points to a GPU hardware fault rather than a software/driver issue.

### **Step 5: Check driver status in the OS**

In Device Manager (Windows) or Activity Monitor/System Report (Mac), check if the GPU is recognized and whether the driver shows an error/warning icon. Reinstall or roll back the GPU driver if issues are found.

### **Step 6: Test the GPU in another system (if possible)**

If available, install the GPU into a different, known-working PC. If it still fails to work there, this confirms the GPU itself is faulty rather than a compatibility/PSU issue with the original system.

### **Step 7: Confirm the fix**

After each fix attempt (reseating, driver reinstall, etc.), reboot and check for stable display output and run a light graphics task (e.g., open a game or benchmark) to confirm the issue no longer occurs.

## **Task 3:**

### **Storage Disconnect/Reconnect Simulation**

**Before disconnecting:** Note the drive's current status in BIOS (usually under "Storage Configuration" or similar) — it should show the drive's model name and capacity.

#### **After unplugging (SATA/NVMe):**

- **BIOS boot screen:** May show "No bootable device found" or simply skip listing that drive in the boot order.
- **BIOS storage menu:** The drive slot will show "Not Detected" or be blank where it previously listed the drive.
- **If it was the boot drive:** The system will likely fail to boot into the OS entirely, possibly showing "Disk Boot Failure, insert system disk and press Enter"

### After reconnecting:

- BIOS should redetect the drive automatically on the next boot — check the storage menu again to confirm it reappears with the correct model/capacity.
- If it's the boot drive, the system should boot normally into the OS again.

## Task 4:

### Random Restarts — Which Component to Check First

#### Check first: RAM

**Reasoning:** RAM issues are the most common cause of random restarts/crashes because faulty or unstable RAM directly corrupts data being actively processed, which the OS or CPU detects as a critical error and triggers an immediate restart as a protective measure. It's also the **easiest and fastest to test** compared to GPU or storage, requiring no special tools — just reseating or a built-in OS memory diagnostic. GPU issues more typically cause crashes specifically during graphics-intensive tasks (not truly random), and storage issues more commonly cause freezing/file corruption rather than sudden restarts — making RAM the most probable and most practical starting point.

#### Diagnostic steps for RAM:

1. Power off, reseal each RAM module firmly in its slot.
2. If multiple sticks are installed, test with **one stick at a time** in different slots to isolate whether a specific module or slot is faulty.
3. Run the **Windows Memory Diagnostic tool** (mdsched.exe), letting it run a full pass to check for memory errors.
4. If errors are found, replace the faulty module; if no errors are found after testing all RAM individually, move on to checking the PSU's voltage rails or the GPU as the next likely suspects.